

VeriLens

Deepfake / AI-Generated Image Detector for KYC

Pre-Presentation Reference Guide

Easy English · Technical Detail · Examples · Honest Limits

One line: VeriLens takes an ID document photo and a live selfie, runs independent forensic lanes over both, and returns three separate explainable verdicts — refusing to guess when evidence is weak, and anchoring the decision on-chain so it cannot be quietly changed later.

Use this PDF before a viva, demo, or pitch. It covers idea, problem, architecture, every lane, the judge, confidence math, UI data flow, training, embeddings, failures, banking use, scaling, and future plans.

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1. Idea & Problem Statement

1.1 The idea (plain English)

Banks and fintech apps ask you to upload an ID card and a selfie. That flow is called **KYC** (Know Your Customer). It used to assume a photo was hard to fake. Today, anyone with a laptop can generate a face that never existed, or paste a new portrait onto a real ID. VeriLens is a mobile + server system that checks both images together, explains why, and says “I don’t know — ask a human” when it should.

1.2 Problem statement (technical)

- **Attack surface:** Remote onboarding (ID + selfie). Deepfakes are cited as ~11% of global fraud (2026); injection-attack volume (feeding a synthetic image where a system trusts the camera) is reported up ~2,665% YoY (iProov).
- **Industry failure mode:** Most detectors emit one confidence number. That hides two different failures: (1) a *real photo of the wrong person* (identity fraud) vs (2) an *AI photo of the right person* (presentation / synthesis fraud).
- **Research gap:** Published detectors over-rely on a **global spectral artifact** left by inpainting (VAE-related shift across the whole frame). Under Inpainting Exchange (INP-X, arXiv 2602.00192), accuracy of commercial detectors can collapse ~91% → ~55% (near chance). The best published fix (FUSED, Aug 2026) explicitly **excludes face manipulation** — which is exactly KYC’s domain.
- **Honesty requirement:** A wrong REJECT locks a real customer out of a bank account. So abstention (INSUFFICIENT_EVIDENCE → REVIEW) must be a first-class output, not an error edge case.

1.3 What we are NOT claiming

We do **not** claim novel forensic algorithms. Error Level Analysis (ELA), noise-residual analysis, and patch classifiers are established. Our contribution is a **KYC-shaped system**: paired ID+selfie, three independent verdict axes, mandatory abstention, signed + on-chain decision record — aimed at a documented blind spot around local face edits.

2. Research We Did

Before writing product code, we mapped what published and commercial detectors actually get wrong for KYC-shaped attacks.

Source	Finding we used	How it shaped VeriLens
arXiv 2602.00192 (INP-X / Jan 2026)	Detectors lean on global artifacts; local edits survive “exchange”.	Lane A trains on exchanged images so it cannot use that shortcut.
FUSED (Aug 2026)	Stronger local-edit detection, but faces excluded from scope.	Opening for a face-centric KYC pipeline.
Wang et al. 2020 (CNNDetection)	JPEG/blur/resize augmentation improves cross-generator generalisation.	Lane A trainer: --augment on by default.
Classic forensics (ELA, noise residuals)	Intra-image inconsistency finds splices without a reference DB.	Lanes B & C are training-free, CPU-only.
iProov / industry fraud reports	Injection & deepfake growth on remote identity.	Selfie is camera-only (no gallery import).
DigiLocker / UIDAI (India context)	Authoritative ID authenticity is a government API problem.	We removed OCR doc-type checks; focus on live selfie forensics.

We also wrote `scripts/verify_baseline.py` to re-measure the commercial ~91%→55% claim on INP-X before quoting it live (vendors may patch after a paper is public).

3. Our Implementation (What We Built)

3.1 Two parts

- **Mobile app (Expo / React Native / TypeScript):** Capture ID + selfie, SHA-256 hash, Ed25519 sign, call forensics API, show 3-axis verdict, anchor digest on Ethereum Sepolia, sync to Supabase, local SQLite offline cache, review queue.
- **Forensics service (Python FastAPI):** Quality gate + forensic lanes + rule-based judge. CPU-first; optional ML weights for Lane A and Lane E.

3.2 Three-axis verdict model

Axis	Values	Meaning
authenticity	REAL / LIKELY_FAKE / INSUFFICIENT_EVIDENCE	Are pixels synthetic or manipulated?
identity	MATCH / MISMATCH / INDETERMINATE / null	Same person on ID and selfie? null = single-image mode
decision	ACCEPT / REJECT / REVIEW	What should ops do?

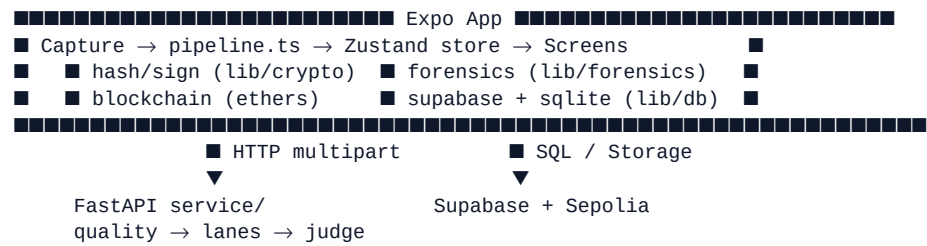
3.3 End-to-end pipeline (app)

1. Hash → SHA-256 of ID bytes + selfie bytes
2. Sign → Ed25519 over the pair (device key in Secure Store)
3. Forensics → POST /v1/analyze (multipart id_image + selfie)
4. Anchor → Sepolia self-transfer with verdict digest calldata
5. Cloud → Supabase case row + optional image upload

Forensics failure is fatal (no verdict to anchor). Anchor/cloud failures are recorded but non-blocking — the local SQLite case still keeps the result.

4. Architecture & Where UI Data Comes From

4.1 High-level diagram



4.2 Where every UI field comes from

UI element	Data source	How it arrives
ID / selfie preview	Device camera or gallery URI	capture.tsx → local file URI
Pipeline step list	lib/pipeline.ts progress callbacks	Hash → Sign → Forensics → Anchor → Cloud
authenticity / identity / decision	service/judge.py → AnalyzeOut.verdict	HTTP JSON from POST /v1/analyze
confidence + “(uncalibrated)”	verdict.confidence + confidence_is_calibrated	UI must show uncalibrated while flag is false
Per-lane score / reasons / box	LaneResult from lanes A/B/C/...	id_image.lanes[] + selfie.lanes[]
Case history / gallery	expo-sqlite (lib/db.ts)	Written after pipeline; offline-first
Review queue	SQLite filter decision=REVIEW	review.tsx + optional Supabase sync
On-chain TX / proof verify	Sepolia + Blockscout API	lib/blockchain.ts; verify-proof screen
Device public key / wallet	expo-secure-store	profile.tsx reads crypto + wallet helpers

Important: The UI does not invent forensic scores. It only displays what the FastAPI service returns (plus local crypto / chain metadata). If the service is unreachable, forensics fails and there is no authenticity guess on the client.

5. Every Lane Explained

Think of lanes as independent “specialist doctors.” Each answers one narrow question. Each can **abstain** if the image does not give enough signal. A separate **judge** combines them with rules (not another neural net).

Quality Gate (Lane Q) — before any forensic claim

Forensic traces live in high-frequency detail. If the image is too small, too blurred, or too heavily recompressed, the gate marks `quality.usable = False` and the judge abstains.

- **min_side_px = 256** — short side below this → unreadable.
- **min_laplacian_var = 60** — variance of Laplacian; low = blur / soft upsample.
- **min_jpeg_quality = 55** — estimated from JPEG quantization tables; heavy recompression destroys ELA and residuals.

Lane A — Local Synthesis (optional, trained)

A **patch-level classifier** trained to catch local AI synthesis / inpainting on faces. Trained especially on INP-X *exchanged* images so it cannot lean on the global artifact shortcut. Needs `requirements-ml.txt` + `weights/lane_a.pt`. Without them, it abstains cleanly. Confidence weight is **capped** at `lane_a_confidence_cap = 0.5` because same-distribution val accuracy does not prove phone-photo generalisation.

Lane B — Noise Residual (training-free)

Bilateral-filter the greyscale image; `residual = |original – denoised|`. Block into 16×16 tiles; compute modified z-scores (Iglewicz–Hoaglin, cutoff 3.5). Flag the **low side only** ($z < -3.5$): diffusion content is often unnaturally noise-free for its detail. Confidence scales with median residual energy — no texture ⇒ nothing to compare.

Lane C — Compression / ELA (training-free)

Recompress at JPEG quality 90; measure per-pixel error level; block and z-score. Flag **both sides** ($|z| > 3.5$): pasted or re-edited regions often have a different compression history. If there is no JPEG history (e.g. PNG), confidence drops (~0.30) instead of inventing an ELA story. Both B and C divide by local gradient energy so they do not just flag “every sky” or “every textured edge.”

Lane D — Capture Attestation

Proves the selfie was captured live in-app: server issues a single-use nonce (`GET /v1/attest/nonce`); device signs nonce `|| sha256(selfie)` with Ed25519; service verifies before applying `attested_bonus = +0.10` to judge confidence. **Absence of attestation never lowers confidence** — almost every genuine photo in the world was never attested by our app.

Lane E — Face Match (optional, embeddings)

Compares ArcFace embeddings from the ID portrait and the selfie (cosine similarity). This is what makes the system “for KYC,” not a generic fake-image filter. See Section 6.

Lane G — Screen / Print Replay (new, capped)

Patch-wise FFT moiré detection (4×4 patches). A real screen replay lights up many patches; a hologram sticker lights up one corner. Confidence capped at 0.4 until labelled calibration exists. Remaining risk: wallpaper / striped shirt filling most of the frame can look “widespread.”

6. Face Embeddings (Lane E)

An **embedding** is a fixed-length vector (here from ArcFace / insightface buffalo_l) that places faces in a space where the same person clusters together. We take the **normed embedding** for each face and compute **cosine similarity** = dot product of two unit vectors (range roughly $-1 \dots 1$; same person tends higher).

- Model input size: **112×112**. Crops smaller than that are upsampled → weaker embeddings.
- Minimum usable face width: **40 px** (below → return None, no guessed score).
- Low-quality margin: if face < 112 px, MATCH needs face_match_above + 0.15 (0.38 → 0.53).
- ID may have a ghost portrait: we pick the **largest** face on the ID.
- Selfie with >1 face → abstain (cannot know which applicant).
- Missing insightface deps → None + reason → identity INDETERMINATE → REVIEW.

Thresholds (config.py)

Similarity s	Identity call
$s \geq 0.38$ (or ≥ 0.53 if low-quality face)	MATCH
$s \leq 0.22$	MISMATCH
otherwise	INDETERMINATE → REVIEW
s is None (no face / no model)	INDETERMINATE on pair checks

Example: Same person, full-res selfie vs tiny Aadhaar crop (~94 px) gave similarities ~0.73 vs ~0.57 in live testing. Both can still MATCH, but the low-quality path exists to stop a barely-over-threshold weak crop from becoming a confident MATCH.

7. How Lane Confidence Is Computed

Each lane returns {score, confidence, usable, reasons, box}. **score** $\in [0,1]$ is “how fake / anomalous” (higher = more suspicious). **confidence** $\in [0,1]$ is “how much this lane trusts its own read.” If confidence < min_lane_confidence (0.35), the lane sets usable=False and is dropped from the judge average.

Shared spatial scoring (Lanes B & C)

- 16×16 non-overlapping blocks → per-block statistic → gradient-normalised → modified z-score vs median/MAD.
- Connected clusters of outlier blocks; ignore clusters smaller than 4 blocks.
- Cluster area → score via saturating curve: ~5% of frame flagged $\approx$ score 0.6.
- Lane B confidence $\uparrow$ with residual energy; Lane C confidence from estimated JPEG quality (or ~ 0.30 if no JPEG history).

Other lanes

- **Lane A:** patch classifier probability / max over face region; weight also tied to checkpoint val_acc_exchanged, then capped at 0.5.
- **Lane D:** does not produce a fake-score; only adds +0.10 to judge confidence when crypto attestation verifies.
- **Lane E:** produces similarity for the identity axis (not the fake score average), though reasons appear in the verdict list.
- **Lane G:** severity $\times$ patch coverage; confidence fixed/capped at 0.4.

Critical honesty flag: confidence_is_calibrated = false. A UI confidence of 0.72 does **not** mean “72% probability fake.” It means usable lanes were fairly confident and agreed. Until a held-out calibration set exists, always say “(uncalibrated).”

8. How the Judge Decides (with Examples)

The judge in `service/judge.py` is **rule-based on purpose**: no honest meta-training set yet, and rules can explain *why*.

8.1 Step-by-step algorithm

1. Compute identity from `face_similarity` (Lane E) if available.
2. If `quality.usable` is `False` → authenticity `INSUFFICIENT_EVIDENCE`;
decision = `REJECT` if `identity==MISMATCH` else `REVIEW`.
3. Keep only lanes with `usable=True` (`confidence ≥ 0.35`).
4. If `usable count < 2` → abstain (need cross-check).
5. `agg = weighted_average(scores, weights=confidences)`
6. `spread = weighted RMS deviation from agg`
7. If `spread > 0.28` → abstain (lanes disagree).
8. `base_conf = mean(weights) * (1 - 0.5*min(spread/0.28, 1))`
+ optional `attested_bonus 0.10` (cap at 1.0)
9. If `agg ≥ 0.65` → `LIKELY_FAKE`; elif `agg ≤ 0.35` → `REAL`;
else abstain (uncertainty band 0.35–0.65).
10. Fold decision:
 `LIKELY_FAKE` → `REJECT`
 else `MISMATCH` → `REJECT`
 else `INDETERMINATE` → `REVIEW`
 else → `ACCEPT`

8.2 Worked examples

Example A — Clean real customer

Usable lanes B=0.05 (conf 0.85), C=0.08 (conf 0.80). face sim=0.71. $agg \approx 0.06 \leq 0.35$ → authenticity `REAL`, identity `MATCH` → **decision ACCEPT**. $base_conf \approx \text{mean}(0.85, 0.80) \times (1 - \text{small spread}) \approx 0.8$.

Example B — AI selfie, matching face

Lanes flag high fake scores (e.g. B=0.82, C=0.70, maybe A=0.90 capped). $agg \geq 0.65$ → `LIKELY_FAKE` → **REJECT** even if identity `MATCH`. This is presentation fraud: right identity claim, synthetic capture.

Example C — Real photo, wrong person

Authenticity lanes look clean (agg low → `REAL`) but face sim=0.12 → `MISMATCH` → **REJECT**. Identity fraud is independent of pixel authenticity.

Example D — Lanes disagree

B score 0.10, C score 0.90, both high confidence. Weighted spread > 0.28 → authenticity `INSUFFICIENT_EVIDENCE`, decision `REVIEW` (unless `MISMATCH`). We refuse to average a fight into false certainty.

Example E — Blurry upload

Laplacian variance < 60 → quality gate fails before aggregation. score=null, confidence=0, `REVIEW`. Better than a confident wrong reject.

Example F — Bug we fixed (MISMATCH + abstain)

Old bug: every abstain path forced decision=`REVIEW` even when Lane E already proved `MISMATCH`. Fixed in `_abstain()`: `MISMATCH` still `REJECTS` when authenticity cannot be decided. Regression: `test_mismatch_rejects_even_when_authenticity_abstains`.

9. Key Design Decisions — What & Why

Decision	Why
Lanes + rule judge (not one black-box model)	Explainability; abstention; no honest meta-train data yet.
Three axes, not one blended score	Identity fraud $\neq$ synthesis fraud; compliance needs the split.
Wide uncertainty band (0.35–0.65)	Wrong REJECT harms real bank applicants.
Selfie camera-only (no gallery)	Blocks classic injection of a pre-made deepfake file.
Attestation only raises confidence	Missing attestation is normal for real photos.
Optional Lane A/E; core runs on B/C	Graceful degradation on free CPU / missing weights.
Cap Lane A confidence at 0.5	High val_acc on same dataset $\neq$ phone-photo generalisation.
Cap Lane G at 0.4	New / unvalidated against labelled replay dataset.
Sign + Sepolia-anchor the verdict digest	Prove the decision wasn't edited later, not only the photo.
Remove OCR document-type lane	DigiLocker/UIDAI already own authoritative ID authenticity.
confidence_is_calibrated=false forever until W5	Don't present raw agreement as a probability.

10. Problems We Ran Into

- **Lane A overfit to curated CelebA-HQ/INP-X:** val_acc_exchanged ~0.99, but real phone photos: confident false positives and a missed full-frame AI headshot. Mitigated with face-crop scan + confidence cap; retrain prepared (140k faces + augmentation).
- **Attestation never worked over HTTP:** FastAPI treated fields as query params beside File(...). Fixed with Form(None); route-level regression test.
- **Lane G scoring bugs (3 rounds):** area-only ignored severe tiny moiré; severity-only over-triggered; whole-image FFT false-positived holograms. Fixed with patch-wise coverage.
- **UI text invisible:** alignItems broke flex text width on reason rows. Fixed at component root width:100%.
- **Groq Lane A refine:** privacy (sends ID/selfie off-device), non-deterministic answers, rate limits; left optional and undisclosed as own lane by owner direction — tension with explainability ethos.
- **Identity MISMATCH swallowed by abstain:** fixed as Example F above.
- **Kaggle trainer wiring:** FACES140K_ROOT wrongly forced when dataset not attached — fixed to shallow-search inputs.

11. Where the Model Fails — Why, Fix, Why Not Fixed Yet

Failure	Why it happens	How to fix	Why not fully fixed yet
Fully synthetic image with uniform stats passes B/C	Intra-image consistency: no local anomaly if whole frame is synthetic.	Strong Lane A / external detector trained on whole-image fakes (140k).	Retrain ready; needs compute + out-of-distribution validation.
Lane A FP/FN on real phones	Train/val from same narrow distribution; domain shift.	Broader data, JPEG/blur augment, drop face-only-only training; then re-test on real photos.	Didn't ship a "pretty" broken model as authority — capped weight instead.
Heavy beauty/denoise selfie	Modern ISP reduces sensor noise → can look "too clean."	Recalibrate thresholds; more phone ISP diversity in eval.	Live tests: B self-excluded via low confidence; no proven bug.
Periodic wallpaper / striped shirt → Lane G FP	Truly widespread high-frequency pattern.	Calibrate on labelled replay set; maybe temporal video cues.	Confidence already capped; labelled dataset not collected.
Low-res ID face embeddings	Upsample to 112px weakens ArcFace vector.	Higher match margin (done); ask better capture UX.	Margin shipped; cannot invent pixels that aren't there.
Uncalibrated confidence	No held-out reliability diagram / temperature scaling.	Build calibration set; set confidence_is_calibrated=true.	Honesty > fake precision; flag stays false on purpose.
Nonce store in-memory	Single process demo; lost on restart / multi-worker.	Redis/DB-backed nonce store.	Enough for demo; production scaling item (see §14).

Safety net observed in testing: Across hard cases this session, min_usable_lanes + disagreement gates meant the system did not emit a false ACCEPT; it preferred REVIEW. That is the intended failure mode.

12. Data Tests & Approximate Training

12.1 Automated tests

- `service/test_service.py` — assert-based, seeded synthetic images: quality gate, abstention paths, Lane B splice catch, attestation never lowers confidence, identity independent of authenticity, MISMATCH-with-abstain regression (~11 core checks historically).
- `test_lane_screen.py` — moiré / localised vs widespread scoring.
- `test_lane_face.py`, `test_lane_a_refine.py`, `test_train_lane_a.py` — optional/ML paths.
- Manual live tests on real phones: real people, ID cards, AI headshots, screen replays, laminated hologram cards.

12.2 Approximate training (Lane A)

- **Primary data:** Inpainting Exchange (INP-X) — face-only CelebA-HQ subset historically ~2010 train / ~893 held-out mask-paired edits.
- **Metric that matters:** `val_acc_exchanged` (setting where published detectors fall to chance). Checkpoint can report ~0.99 on that split — *not* proof of real-world readiness.
- **Prepared retrain mix:** INP-X + xhlulu/140k-real-and-fake-faces (~70k real FFHQ + ~70k StyleGAN fakes) + CnnDetection-style augment (random JPEG / blur / resize). Prefer keeping non-face INP-X domains for diversity (not only --face-only).
- **New headline metrics on checkpoint:** `val_acc_faces140k_real` / `_fake` alongside `exchanged` / `inpainted` / `original`.
- **Hardware:** runs on Kaggle T4, Colab, or local (e.g. Apple MPS).
- **Rule:** before trusting a new `lane_a.pt`, re-test on real non-dataset photos — not only the training distribution's held-out split.

14. How We Scale to Production

Layer	Demo today	Production scale
Forensics compute	Single uvicorn, CPU	Horizontally scaled workers behind a queue (Redis/SQS); GPU pool for Lane A/E; autoscaling; request timeouts + circuit breakers.
Attestation nonces	In-memory dict	Redis/DB with TTL; sticky-free multi-instance.
Model serving	Lazy load in process	Dedicated inference service; batch faces; versioned model registry.
App backend data	SQLite + Supabase free	Managed Postgres; object storage; RLS per tenant/bank; PII encryption.
Blockchain anchor	Sepolia testnet self-tx	Mainnet or permissioned ledger; batch Merkle roots to cut gas; or hash-anchoring service.
Calibration	Uncalibrated thresholds	Bank-specific holdout; monitoring for drift; human feedback loop from REVIEW.
Privacy	Optional Groq refine	Default on-prem / VPC only; no third-party image egress without consent.
SLA	Best effort demo	p99 latency budgets; fallback REVIEW if forensics times out (never silent ACCEPT).
Security	Device Ed25519 + HTTPS	mTLS to bank VPC; key rotation; SOC2 logging; red-team injection tests.

Scaling principle: when in doubt under load or model failure, **route to REVIEW**, never invent ACCEPT.

15. Future Plans

- Finish Lane A retrain on INP-X + 140k faces + augmentation; validate on real phone photos before raising confidence cap.
- Build a true calibration set → reliability diagrams → flip confidence_is_calibrated when earned.
- Production nonce store + multi-worker FastAPI deployment.
- Bank pilot: integrate as pre-screen webhook into existing KYC vendor.
- Stronger liveness (challenge-response / short video) beyond single-frame attestation.
- Tenant isolation, consent flows, and remove/disable third-party LLM refine for regulated PII.
- Optional mainnet / permissioned anchoring with batched Merkle proofs.
- Continuous evaluation harness against new generators (as diffusion models evolve).

16. Quick Q&A Cheat Sheet

“Isn’t this just another deepfake detector?” — No. KYC is a *paired identity* problem. Without ID↔selfie match and honest abstention, it’s a filter with a KYC sticker.

“What’s novel?” — System design for KYC + documented local-edit gap for faces — not a new named algorithm. Say that plainly.

“Walk me through an upload.” — Quality gate → lanes in parallel → judge gates (usable count, disagreement, band) → 3-axis verdict + reasons → hash/sign/anchor.

“What if a lane is missing?” — It abstains; judge continues if ≥ 2 usable lanes remain; otherwise REVIEW.

“Can confidence be trusted as probability?” — Not yet. `confidence_is_calibrated` is false by design.

“Where does the UI get numbers?” — Only from the FastAPI JSON (plus local hash/signature/tx metadata). Client never invents forensic scores.

“In the age of AI, truth needs a receipt.” — VeriLens

Companion docs in repo: `README.md` · `docs/PROJECT_STORY.md` · `HANDOFF.md` · `DEMO_SCRIPT.md` · `service/README.md` · `pitch/VeriLens_Pitch.pptx`